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Hydraulic Erosion Engine – Real-Time Procedural Terrain Deformation System

Hydraulic Erosion System Overview

This module transitions terrain from a static hydrological analysis structure into a fully dynamic physical simulation system. Terrain is no longer only interpreted for water behavior; it is actively reshaped by hydraulic forces over time.

<https://www.udemy.com/course/web-based-3d-terrain-industrial-printing-engine-mastery/>

System Position in Engine Architecture

Earlier modules focused on terrain generation and basin detection. This stage introduces active deformation, where water becomes a transformation mechanism that continuously modifies the terrain surface.

Core Objective: Terrain Evolution Simulation

The primary goal is to simulate how water reshapes terrain over time through erosion, sediment transport, and deposition. This transforms the system into a time-dependent geological simulation rather than a static model.

Water as a Geometric Transformation Agent

Water is modeled as an active system that removes, moves, and deposits material across the terrain surface. This creates a distributed geometric editing process where elevation is continuously updated based on physical flow behavior.

Flow Direction Vector Field

while slope intensity determines flow velocity.

Sediment Transport System

Sediment acts as a transferable mass within the system. Erosion removes material from high-energy regions, while deposition deposits it in lower-energy zones. This creates a closed redistribution system where terrain is reshaped rather than created or destroyed.

Terrain Carving Mechanism

As sediment moves across the terrain, it gradually reshapes surface geometry. High elevations erode, valleys deepen, and river channels form naturally through continuous material redistribution.

Iterative Erosion Loop System

Hydraulic erosion operates as a time-based simulation loop. Each iteration computes flow direction, simulates water movement, applies sediment transport, and updates terrain height, producing progressive geological evolution.

Full Erosion Pipeline Architecture

The pipeline includes heightmap input, flow vector computation, velocity simulation, sediment transport, terrain updating, and iterative feedback loops. Each cycle modifies the input state of the next iteration.

System Mental Model

Terrain is treated as a deformable physical substrate. Water functions as a distributed force field, and sediment represents mass exchange. Erosion becomes a continuous geometric rewriting process over time.

Engine Behavior Over Time

Initial terrain features appear artificial and sharp. After multiple erosion cycles, terrain evolves into natural formations with smooth mountains, river valleys, and realistic drainage networks.

GPU Computational Strategy

The system is optimized for parallel execution using heightmap textures. Each cell is processed independently using compute shaders, enabling real-time erosion simulation across large terrains.

The model approximates natural geological processes such as rainfall runoff, surface erosion, sediment transport, and river formation, making it a simplified but physically grounded simulation system.

Engineering Transformation Summary

The system evolves from a static terrain model into a dynamic simulation engine. It introduces flow computation, sediment transport, iterative deformation loops, and GPU-based physical terrain evolution.

Student Questions

Why is hydraulic erosion a feedback-based system

How does gradient determine water flow direction

What role does sediment play in terrain transformation

Why are iterative loops necessary for realistic erosion

How does GPU parallelism enable real-time simulation

Student Assignments

Implement a flow vector field based on terrain gradients

Build a sediment transport simulation system

Create an iterative erosion loop for terrain evolution

Design a terrain carving system with mass redistribution logic

Explain the GPU-based architecture of erosion simulation

Final Engine Statement

At this stage, terrain is no longer a static dataset. It becomes a continuously evolving physical system driven by flow dynamics, sediment transport, and iterative deformation loops, forming a real-time geological simulation engine running inside the browser.

